

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read grayscale image
image = cv2.imread('grayscale.jpg', cv2.IMREAD_GRAYSCALE)

# Check if image is loaded
if image is None:
    print("Error: Image not found!")
else:

    # 8-bit image (256 gray levels)
    original = image.copy()

    # 6-bit quantization (64 gray levels)
    quant_64 = (image // 4) * 4

    # 4-bit quantization (16 gray levels)
    quant_16 = (image // 16) * 16

    # 2-bit quantization (4 gray levels)
    quant_4 = (image // 64) * 64
```

```
# Display results
plt.figure(figsize=(16,5))

plt.subplot(1,4,1)
plt.imshow(original, cmap='gray')
plt.title("Original (256 Levels)")
plt.axis('off')

plt.subplot(1,4,2)
plt.imshow(quant_64, cmap='gray')
plt.title("64 Gray Levels")
plt.axis('off')

plt.subplot(1,4,3)
plt.imshow(quant_16, cmap='gray')
plt.title("16 Gray Levels")
plt.axis('off')

plt.subplot(1,4,4)
plt.imshow(quant_4, cmap='gray')
plt.title("4 Gray Levels")
plt.axis('off')

plt.show()
```

Original (256 Levels)



64 Gray Levels



16 Gray Levels



4 Gray Levels



